

Transonic Flowfield in a Supersonic Nozzle with Small Throat Radius of Curvature

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Nomenclature

M = Mach number
 p = static pressure, psia
 p_t = total pressure, psia
 r = radius, in.
 r_c = throat wall radius of curvature, in.
 r_{th} = wall radius at throat, in.
 T_t = stagnation temperature, °R
 z = axial distance from throat, in.
 γ = specific heat ratio

Introduction

THE important parameter that governs the isentropic flowfield in the transonic region of a supersonic nozzle with a circular-arc throat is the ratio of throat radius of curvature to throat radius, r_c/r_{th} . For values of r_c/r_{th} considerably greater than unity, i.e., gradual throat contour, the flow is essentially one-dimensional. Two-dimensional flow effects become important as the ratio r_c/r_{th} decreases. Provided that r_c/r_{th} is not less than about 2, existing two-dimensional flow theories¹⁻³ adequately predict the transonic flowfield as indicated by the wall static pressure measurements.⁴ However, for nozzles with tighter throats ($r_c/r_{th} < 1$), such as are found in some rocket engines, these theories do not apply. This Note presents some internal flow measurements in the transonic region of a nozzle with a small ratio of r_c/r_{th} (0.625) and compares some recently developed prediction methods of other investigators with the data.

Experimental Apparatus

The conical nozzle used in these experiments had half-angles of convergence and divergence of 45° and 15°, respectively. The static pressure in the flowfield was measured within ± 0.2 psi through a 0.006-in.-diam hole drilled radially through the wall of a 0.035-in.-diam tube that extended axially through the nozzle. The tube, which was under tension to hold it taut, was remotely supported in a plenum chamber upstream and in a vacuum chamber downstream. Axial traverses were made along the centerline and at five radial locations in the nozzle throat region. The incremental axial distances traversed and the locations of the wall static pressure taps were measured within 0.002 in. However, the coordinates of the static pressure hole were known only within 0.01 in.

The tests were conducted with air at a stagnation pressure of 70 psia and a stagnation temperature of 540°R. The boundary layer 1.4 in. upstream of the nozzle inlet was turbulent and had a thickness of about 5% of the duct radius. The momentum thickness Reynolds number was 2100 as deduced from the velocity distribution measured with a 0.004-in.-high, flattened-tip pitot probe. The actual boundary-

layer thickness in the throat region is not known but, from boundary-layer calculations, was considerably less than 0.070 in., the nearest distance from the wall that the static pressure was measured with the traversing tube.

Experimental Results

Axial distributions of internal static pressure measurements are shown at various radial locations in Fig. 1. The radial static pressure variation is large, with values of the static pressure along the centerline as much as three times those at the wall. In fact, the radial pressure variation is so large that near the wall at the throat, the radial static-pressure gradient is about the same as the axial static-pressure gradient. The gas expands more rapidly along the wall than along the centerline, and there are correspondingly large radial variations in the Mach number (Fig. 1) calculated for isentropic flow ($\gamma = 1.4$). In the inviscid flowfield in the throat plane, the Mach number is 0.8 at the axis and 1.4 at the edge of the boundary layer.

The measured static-pressure rise along the wall just downstream of the tangency between the circular-arc throat and the conical divergent section shown in Fig. 1 is believed to be associated with a compressive turning of the flow.⁵ The persistence of the strong angular motion acquired by the flow in the small-radius-of-curvature throat region can lead

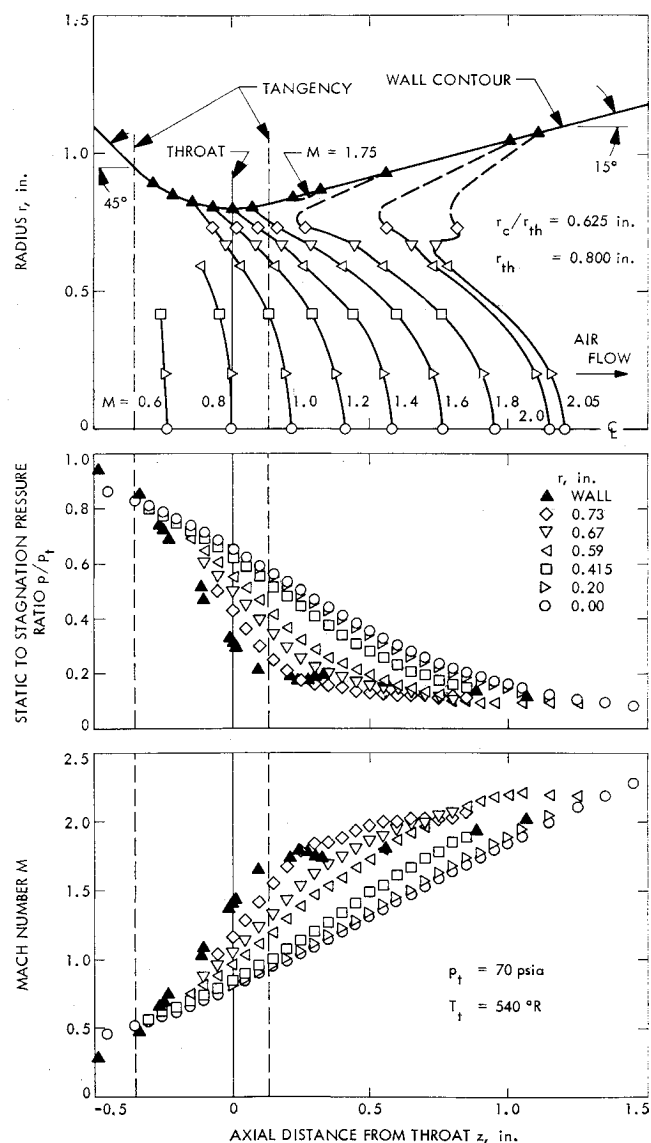


Fig. 1 Experimentally measured static pressure and calculated Mach number distributions in the transonic region.

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to an overturning of the flow so that streamlines near the wall become inclined to the downstream conical wall, as indicated by calculations. The region where Mach lines begin to converge, which is the onset of shock formation associated with the compressive turning of the flow to become parallel to the wall, is evident in Fig. 1 from the shape of the Mach number contours. The weak oblique shock wave forms subsequently and extends downstream, intersecting the centerline⁶ at $z = 3.15$ in. It is reasonable to assume that the flow is isentropic throughout this region.⁶

When the flow is parallel to the tube, the true static pressure is measured. At some off-axis locations, the flow is inclined to the tube. At these locations the orientation of the pressure tap relative to the velocity component normal to the tube was either about 110° for convergent flow or about 80° for divergent flow. Since the pressure tap was on the side of the tube, the pressure measured was lower than the true static pressure⁷ and a higher-than-true-Mach number was calculated. Such measurements in the subsonic region where a significant error would be expected are not shown. In the region downstream of the throat where the onset of shock formation is evident in Fig. 1, all data were reduced to show the trends observed even though errors are expected. The true location of the Mach number contours in this region would be further downstream and distorted as indicated by the shapes of the curves shown there.

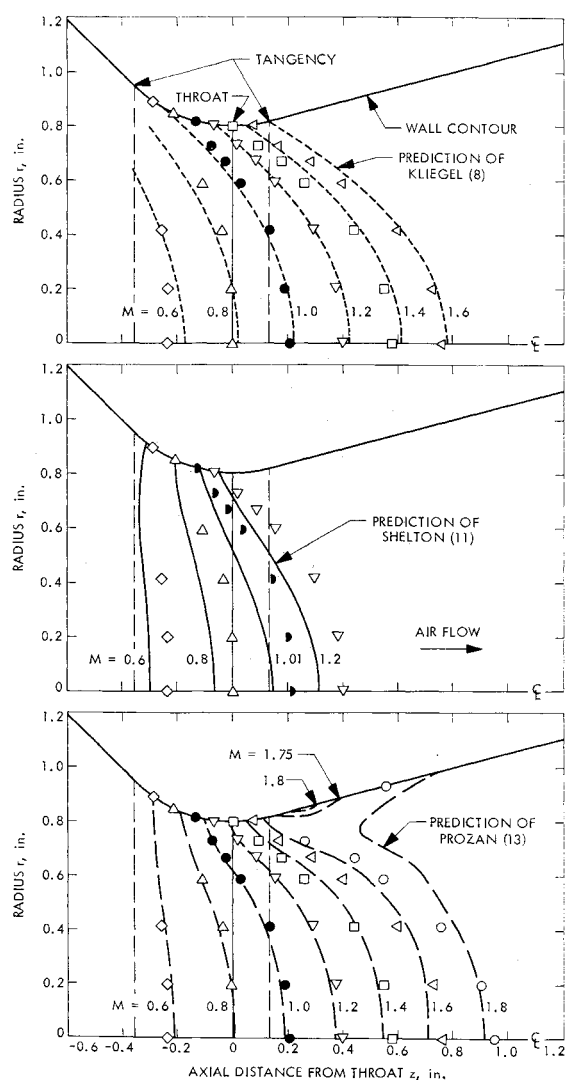


Fig. 2 Comparison of theoretical predictions with observed Mach number distributions in the transonic region. Symbols represent author's data from Fig. 1.

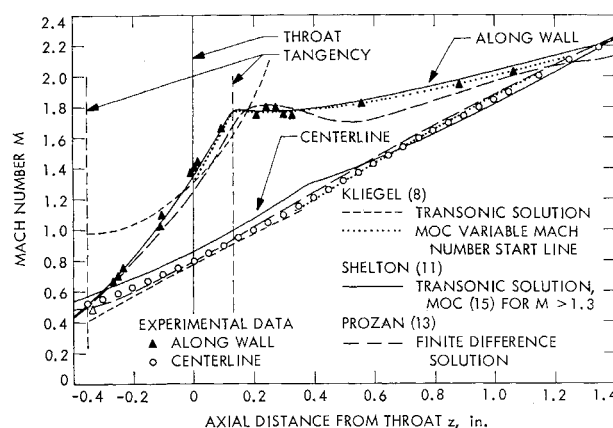


Fig. 3 Comparison of wall and centerline Mach number distributions, from transonic and method-of-characteristics (MOC) supersonic flow predictions, with the experimental results.

Evaluation of Theoretical Predictions

Recently developed approximate methods which may be used for predicting the flow through the transonic region of such a tight throat nozzle can now be evaluated as a consequence of the internal flow measurements. Also of interest is the agreement of the flowfield farther downstream with predictions using the method of characteristics based on starting conditions obtained from the transonic flow solutions. A brief discussion of three methods for inviscid, adiabatic flow of a perfect gas with $\gamma = \text{const}$ follows.

Kliegel and Levine,⁸ using a series expansion method about the sonic condition similar to that of Hall,³ sought a transonic solution for the velocity field in toroidal coordinates for nozzles and noted that using the wall coordinate as a normalization quantity introduced the perturbation parameter $1/[1 + (r_c/r_{th})]$. Therefore, Hall's result for an expansion in $1/(r_c/r_{th})$ was transformed into an expansion in $1/[1 + (r_c/r_{th})]$. Kliegel's prediction, using three terms in the expansion, is shown in Fig. 2. The agreement with the experimental data is good in the throat plane, and reasonably good along the centerline, but deviates to a greater extent along the wall as is also seen in Fig. 3. The calculated flow coefficient of 0.982, defined as the ratio of the inviscid, adiabatic, two-dimensional to the one-dimensional mass flow rate, agrees well with an average value for a number of tests of the ratio of the measured mass flow rate to that for one-dimensional, inviscid, adiabatic flow of 0.985.

In a second method, the singularity in the equations of motion at the sonic condition if the nozzle contour were specified is avoided by prescribing the velocity (or pressure) distribution along the centerline and then adjusting this distribution until a particular calculated streamline coincides with the nozzle wall contour. Hopkins and Hill,⁹ using Friedrichs'¹⁰ formulation, applied this inverse method by empirically fitting the particular streamline to the nozzle wall contour, using a prescribed initial exponential velocity distribution along the centerline. However, for the nozzle considered, the fit was inaccurate upstream of the throat. Shelton,¹¹ also using Friedrichs' formulation, has obtained a good streamline fit to the nozzle wall contour to 1.5 throat radii upstream of the throat by choosing a third-order polynomial expression for the centerline velocity distribution and by successively re-evaluating the polynomial coefficients. However, farther upstream the predicted contour deviates from the nozzle wall. This method agrees with the measurements along the wall in the transonic region (Fig. 2) but higher Mach numbers are predicted than are observed in the internal flow.

In the third method, an interesting technique suggested by Crocco¹² was applied to inviscid, adiabatic nozzle flow

by numerically solving the time-dependent equations of motion (hyperbolic in form) in the subsonic, transonic, and supersonic flow regions. The calculation was carried out by Prozan¹³ and information on the technique is described by Saunders.¹⁴ Starting from an initially prescribed flowfield taken to be one-dimensional, the steady-state solution that is sought is obtained as a limit of the solution of the non-steady problem. Boundary conditions must be prescribed around the entire flowfield, and this requires an a priori specification of the mass flow rate, a quantity that actually, however, is to be calculated. This prediction yields an inviscid flow coefficient of 0.990, which is higher than the experimentally measured value of 0.985; but the resulting Mach number distribution agrees generally with the experimental results in the transonic region (Fig. 2). Of particular interest is the similarity between the experimental (Fig. 1) and the predicted (Fig. 2) Mach number contours in the shock-formation region. Along the wall the predicted values (Fig. 3) are somewhat low just downstream of the sonic condition but agree with the data farther downstream, where the Mach number peaks and then decreases slightly. The predicted decrease in the Mach number (wall pressure rise) just downstream of the tangency, however, is larger than that determined experimentally. The pressure rise is important since, under certain conditions, this adverse pressure gradient can significantly influence the boundary layer and, thus, the heat transfer from high-temperature gas flows.⁵ Prozan's prediction agrees well with the experimental data along the centerline.

In two method-of-characteristics predictions shown in Fig. 3, good agreement is found with the experimental results, and a slight Mach number decrease is predicted approximately where observed. Kliegel used a variable-Mach-number start line that varied from $M = 1.3$ along the wall to $M = 1.1$ along the centerline; Shelton used a start line of $M = 1.3$ and the program of Butler.¹⁵

Summary and Conclusions

Internal flow measurements in a nozzle with a small ratio of r_c/r_{th} of 0.625 revealed radial variations in the flow. At the physical throat the Mach number was 0.8 at the axis and 1.4 near the wall. The three methods considered predict the transonic flowfield reasonably well in nozzles which have a small throat radius of curvature ($r_c/r_{th} < 1.0$), such as that for the nozzle investigated.

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Some Aspects of the Natural Coordinate System in the Finite-Element Method

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1. Introduction

EXTENSIVE use of linear, triangular, and tetrahedral elements in the finite-element method have given rise to the search for an appropriate coordinate system for their analysis. Among the various systems proposed, the natural coordinates¹ or the area coordinates,² as they are sometimes called for the particular case of a triangle, have proven to be particularly attractive because of their inherent symmetry with respect to the element. In topology³ the corresponding N -dimensional coordinates are commonly termed barycentric coordinates and are associated with an N -dimensional simplex which in the particular case of a physical space is reduced to the vertices of the element. Here we present a systematic and unified derivation of the basic properties of the natural system and some useful formulas for the analysis of straight or curved elements. Although some of the derived formulas are known, some are believed to be new. Recently, Stricklin⁴ suggested an integration formula for the general polynomial term expressed in area coordinates. A derivation of the corresponding more general formula is presented in this Note.

2. Linear Coordinate System

A point inside (outside) a linear, triangular, tetrahedral, or in general any simplex element is determined by the coordinates

$$\zeta_i, \quad i = 1, 2, \dots, n \quad (1)$$

where n is the number of vertices per element. They are related to the Cartesian system x_i , $i = 1, 2, \dots, N$, by

$$x_i = X_{ij}\zeta_j, \quad i = 1, 2, \dots, N, \quad j = 1, 2, \dots, n \quad (2)$$

in which the matrix X_{ij} collects the coordinates x_i of the vertex j and summation is performed over repeated indices. Assuming that the linear, triangular, and tetrahedral elements are situated in a one-, two-, and three-dimensional

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